

Take-Home Final: TWFE and Modern DiD with staggered adoption)

Assigned: Mon, March 2 \ **Due:** Fri, March 6 (11:59pm)

Overview

This take-home final is an applied differences-in-differences exercise with panel data. You will:

- 1) estimate a two-way fixed effects (TWFE) model,
- 2) build an event-study plot, and
- 3) re-estimate the dynamic effects using a modern estimator designed for staggered adoption.

General instructions

- Submit a compiled PDF *and* your reproducible code (.R/.Rmd).
- Include a brief note describing how you used AI tools (if at all), consistent with the course AI policy.
- Unless a problem asks for a specific format, write short answers and focus on interpretation.

Packages

You may use any R packages you want. If you use specialized DiD packages (e.g., Callaway–Sant’Anna’s `did`), briefly describe what comparison group they use (never-treated vs not-yet-treated) and what estimand your plot represents.

Data

We will use the `Fatalities` panel dataset from `AER` (48 U.S. states, 1982–1988). The policy is the minimum legal drinking age (MLDA), recorded as `drinkage`.

We will focus on traffic fatality rates among 18–20 year olds.

Setup (run this first)

```
library(AER)
library(dplyr)
library(ggplot2)
library(fixest)

data("Fatalities")

df <- Fatalities |>
  mutate(
    year = as.integer(as.character(year)),
    state = as.character(state),
    # Outcome: fatalities per 100,000 people age 18-20
    y = 100000 * fatal1820 / pop1820,
```

```

logy = log(y),
# Treatment indicator: MLDA == 21
D = as.integer(drinkage == 21)
) |>
group_by(state) |>
mutate(
  # First-treated year (cohort); Inf for never-treated states
  G = if (any(D == 1)) min(year[D == 1]) else Inf,
  ever = as.integer(is.finite(G)),
  # For convenience: define event_time for never-treated states as 0
  event_time = ifelse(is.finite(G), year - G, 0L)
) |>
ungroup()

```

Problems

1) Treatment timing + descriptive plots

1.1 Create a table showing how many states have first-treated year $G \in \{1982, \dots, 1988\}$, plus how many are never treated ($G = \infty$).

Deliverable: one small table + brief interpretation.

1.2 Make a plot showing treatment timing by state and year (e.g., a heatmap of D_{it} over time). Order states on the y-axis by their first-treated year G .

Deliverable: one figure + brief description what you see.

1.3 Plot the average outcome y_{it} over time for states that are treated in a given year (i.e., $D_{it} = 1$) vs untreated that year (i.e., $D_{it} = 0$).

Deliverable: one figure.

2) Baseline TWFE (levels and inference)

Let $Y_{it} = \log(y_{it})$.

2.1 Estimate the TWFE model

$$Y_{it} = \alpha_i + \lambda_t + \tau D_{it} + \varepsilon_{it}.$$

Use state and year fixed effects and **cluster standard errors by state**.

Deliverable: report $\hat{\tau}$, a 95% CI, and interpret the magnitude in percent terms.

2.2 Repeat 2.1 but report **i.i.d. (non-clustered) standard errors**. Compare the i.i.d. SE to the clustered SE and briefly explain which is more credible here and why.

Deliverable: one short paragraph.

2.3 Add controls X_{it} of your choice (at least three) and re-estimate the TWFE model. Suggested controls (choose at least three): `income`, `unemp`, `beertax`, `spirits`, `miles`, `emppop`.

Deliverable: one regression table (baseline + controls) and briefly compare the estimates.

3) Event studies: TWFE vs a modern approach

This section asks you to estimate dynamic effects around adoption.

Important sample note: States with $G = 1982$ are treated from the first year we observe, so they have no pre-treatment period in the sample. For event studies, drop states with $G = 1982$.

```
df_es <- df |> filter(G > min(year) | !is.finite(G))
```

3.1 Naive TWFE event-study regression. Estimate an event-study TWFE regression using event time $k = t - G_i$:

$$Y_{it} = \alpha_i + \lambda_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{t - G_i = k\} \cdot \mathbf{1}\{G_i < \infty\} + \varepsilon_{it},$$

with $k = -1$ as the omitted baseline. Cluster by state.

Hint (one way): use `fixest::i(event_time, ever, ref = -1)` on `df_es`.

Deliverable: an event-study plot (leads + lags) + 5–10 sentences interpreting (i) pre-trends and (ii) post-treatment dynamics.

3.2 Modern event-study estimator for staggered adoption. Re-estimate dynamic effects using the Sun & Abraham (2021) interaction-weighted approach, implemented in `fixest` via `sunab(G, year)` in a TWFE regression.

Deliverable: an event-study plot + 1–3 sentences comparing this plot to the naive TWFE plot in 3.1.

3.3 Short answer: Explain (conceptually) why the naive TWFE event-study coefficients can be hard to interpret under staggered adoption with heterogeneous effects.

3.4 (Optional, *fct* replication) The `fct` package (Xu, 2024) provides a unified interface (`did_wrapper`) for estimating TWFE event studies and modern DiD estimators (including interaction-weighted and Callaway–Sant’Anna style estimators), plus built-in event-study plotting (`esplot`).

If you choose this option, do the following:

- 1) Install/load `fct` (once):

```
install.packages("fct")
library(fct)
```

- 2) Estimate event studies with at least **two** methods using `df_es`:

- TWFE (baseline): `method = "twfe"`
- Modern: choose one or both of `method = "iw"` (interaction-weighted) and `method = "cs_notyet"` (not-yet-treated comparisons)

```
# Some methods (e.g., Callaway--Sant'Anna via {did}) require a numeric unit id.
df_es_fct <- df_es |>
  mutate(state_id = as.integer(factor(state)))
```

- 3) Plot the event studies with `esplot(...)` and compare the results to your `fixest` plots in 3.1–3.2.

Note: If you use this option, add a short sentence explaining what control group the method uses (e.g., never-treated vs not-yet-treated).